UNIVERSITY OF CANBERRA

AI-Powered Camera–Radar Fusion for Advanced 3D Object Detection and Road Safety

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1 Introduction

This report introduces an AI-driven camera-radar fusion framework designed to enhance 3D object detection for road safety. While cameras provide rich semantic data, their performance degrades in poor light, bad weather, and at long ranges. Radar offers reliable range and velocity data regardless of visibility but lacks detailed semantics. By fusing these complementary modalities, the framework achieves more reliable and accurate detection of road users than single-sensor systems.

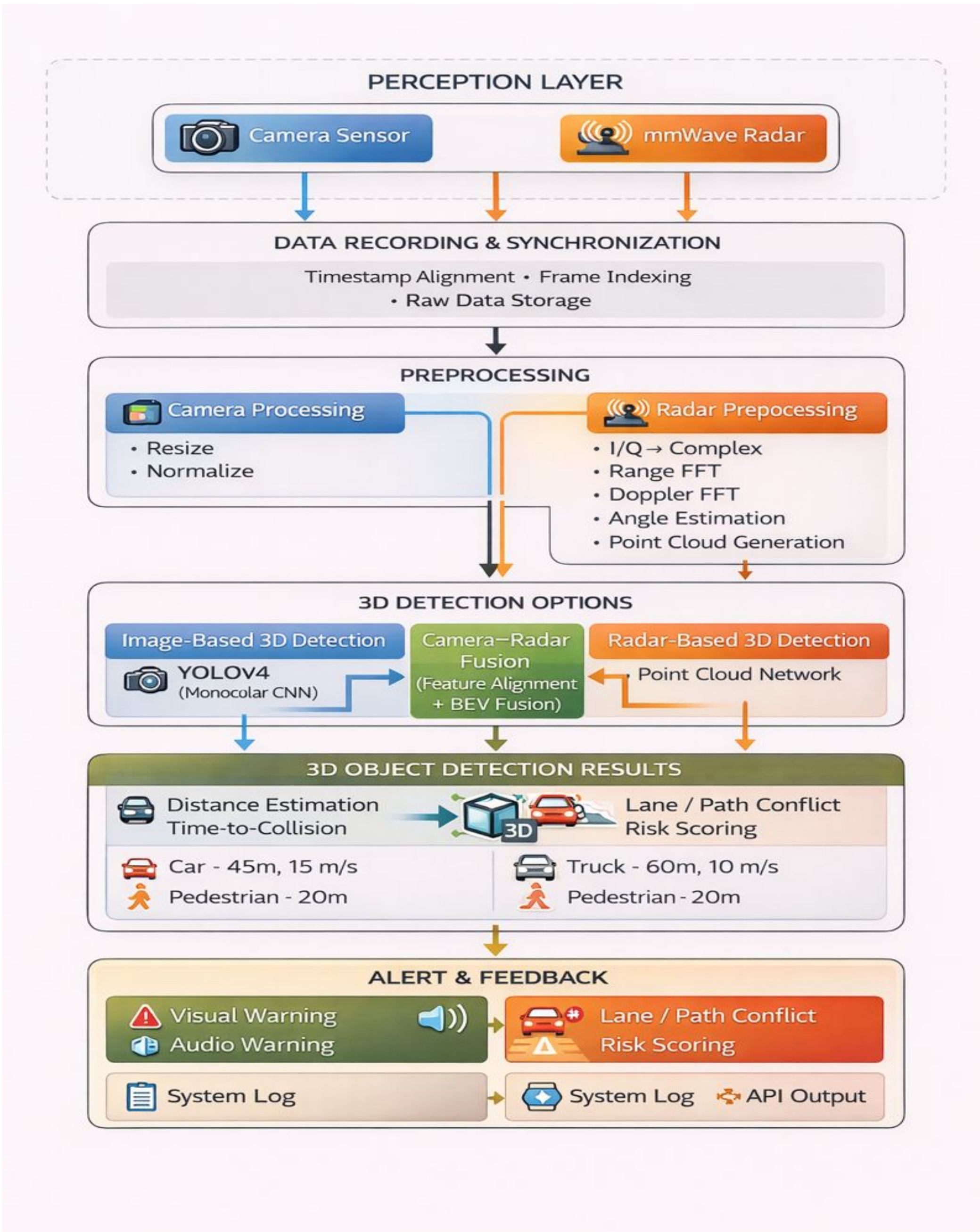


Figure 1: Workflow: end-to-end camera–radar perception and fusion pipeline for 3D object detection and road-safety risk analysis.

This report also describes the design and implementation of a real-time data recording and monitoring system based on a web-based client–server architecture deployed on a mobile vehicle/scooter platform. The framework supports synchronized camera and radar data acquisition, live monitoring, and replay-based analysis through a browser-accessible interface running on an embedded NVIDIA computing board. A full high-resolution radar processing pipeline is detailed, converting raw interleaved I/Q ADC data into 3D point cloud representations using range, Doppler, and angular estimation. Finally, this report discusses risk dimensions and outlines future work focused on advanced camera–radar fusion models, cascaded radar and stereo camera integration, and the creation of a comprehensive, well-documented radar dataset with standardized configurations and preprocessing pipelines to support reproducible research and reliable benchmarking for 3D perception and road safety applications.

The workflow of the framework main components illustrated in Figure 1 that presents an end-to-end camera–radar perception and fusion pipeline for real-time 3D object detection and road-safety analysis. The framework integrates synchronized camera and mmWave radar sensing, followed by data recording and modality-specific preprocessing. Three detection strategies are supported: image-based 3D detection, radar-based 3D detection, and camera–radar fusion using feature alignment and bird's-eye-view (BEV) fusion. The resulting 3D object detection outputs provide object class, position, distance, and velocity, which are subsequently used for risk analysis, including time-to-collision estimation and lane or path conflict assessment. The final stage generates safety-critical alerts and system feedback through visual and audio warnings, system logs, and API outputs.

The **Perception/Sensing Layer** integrates a monocular camera and an mmWave radar sensor mounted on a mobile scooter/vehicle platform, providing complementary visual and geometric measurements. These sensor streams are processed by the **Data Recording and Synchronization** module, which performs timestamp alignment, frame indexing, and raw data storage to ensure synchronized multimodal acquisition and full data traceability. This module is consistent with the report's web-based client–server architecture and enables reliable offline analysis and reproducible experimentation. In the **Preprocessing** stage, camera frames undergo resizing and normalization, while radar signals are converted from interleaved I/Q samples into complex-valued representations, followed by range FFT, Doppler FFT, angular estimation, and 3D point-cloud generation. These operations produce physically meaningful spatial features that are suitable for downstream AI-based detection models.

The **3D Detection Options** layer supports three complementary strategies: image-based 3D detection using a monocular CNN model (e.g., YOLOv4), radar-based 3D detection using point-cloud networks, and camera–radar fusion based on feature alignment and bird's-eye-view (BEV) fusion. The outputs of these branches are consolidated into the **3D Object Detection Results** layer, which provides a structured list of detected objects along with their attributes, including object class, position, distance, and velocity (e.g., car at 45 m and 15 m/s, truck at 60 m and 10 m/s, pedestrian at 20 m). These detection results are then forwarded to the **Risk Analysis** stage, where distance estimation, time-to-collision computation, trajectory reasoning, and lane or path conflict scoring are performed. Finally, the **Alert and Feedback** module converts risk-aware perception outputs into actionable responses, including visual warnings, audio alerts, system logs, and API outputs. Collectively, this pipeline realizes a robust and interpretable camera–radar fusion framework that transforms raw multimodal sensor data into safety-critical information for intelligent mobility and road-safety applications.

Code Repository and Demonstration Videos. The complete implementation of the proposed recording, preprocessing, and perception framework is publicly available as an open-source repository at Source Code. This repository serves as the reference software framework for data acquisition, synchronization, preprocessing, visualization, and replay-based analysis. Supplementary demonstration videos are provided to support reproducibility. A replay of the recording system is available at Recording System Replay Video, and a comparative visualization of camera- and radar-based detections is available at Camera vs. Radar Detection Comparison Video. These resources enable transparent validation and benchmarking of camera–radar based 3D perception methods for road safety applications.

2 Data Recording and Real-Time Monitoring System

This section describes the architecture and operational workflow of the **real-time data recording and monitoring system** developed for road sensing and safety. The system supports live streaming, synchronized data acquisition, and post-experiment analysis through a web-based interface. It follows a **web-based client–server architecture** optimized for deployment on a mobile scooter/vehicle platform. An embedded NVIDIA computing board mounted on the vehicle hosts the server application and provides a browser-accessible graphical user interface for remote control, visualization, and data recording and management. At the hardware level, the system integrates an **mmWave radar sensor** interfaced with a data capture card to support high-bandwidth radar data acquisition, along with an integrated camera to enable multimodal perception. The sensing hardware communicates with the embedded platform via Ethernet for raw data streaming and USB for radar configuration and control. Figure 2 outlines the client-server architecture for the system. The Vehicle Platform Layer handles sensing and streams raw interleaved in-phase (I) and quadrature (Q) and RGB data to the Server Layer, where it is processed via a preprocessing pipeline and managed by a Flask Orchestrator. The processed results are then delivered to the Web Browser Interface in the Client Layer, while a separate Data Flow & Storage Layer records all data locally.

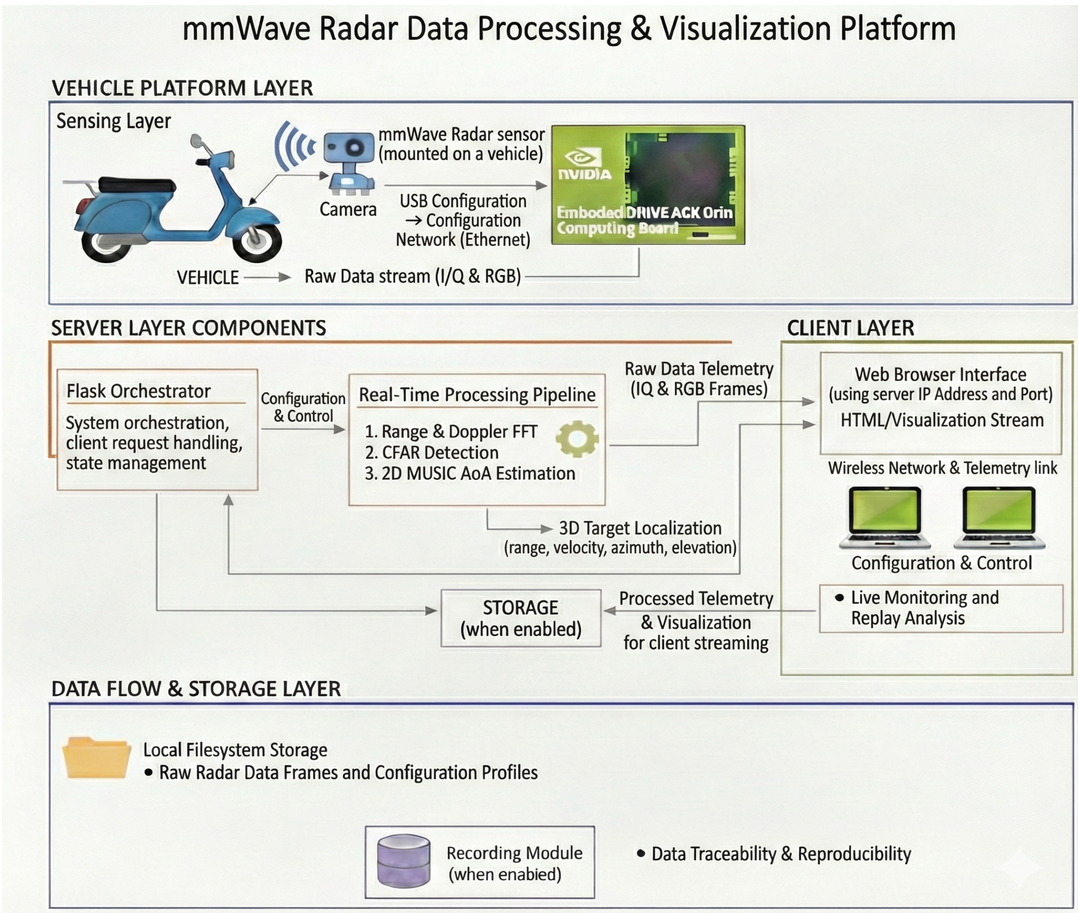


Figure 2: Client-Server Architecture of the mmWave Radar Data Processing & Visualization

The server layer is responsible for system orchestration, data processing, and client communication. A Flask-based web server manages system states and handles client requests, while a real-time signal processing pipeline performs range and Doppler FFTs, CFAR detection, and 2D MUSIC-based Angle of Arrival estimation for three-dimensional object detection. The client layer consists of a standard web browser that provides platform-independent access to the system through the IP address and port of the embedded board. The interface supports a live mode for real-time monitoring and a replay mode for post-experiment visualization and analysis. During operation, the system is initialized through the web interface, after which the radar sensor and the cameras stream raw IQ and RGB samples to the server. Incoming data are processed in real time, and the resulting telemetry and visualization output are streamed back to the client for road monitoring. When recording is enabled, the server stores raw radar data frames and their associated configuration profiles on the local file system, ensuring data traceability and reproducibility throughout real-world testing.

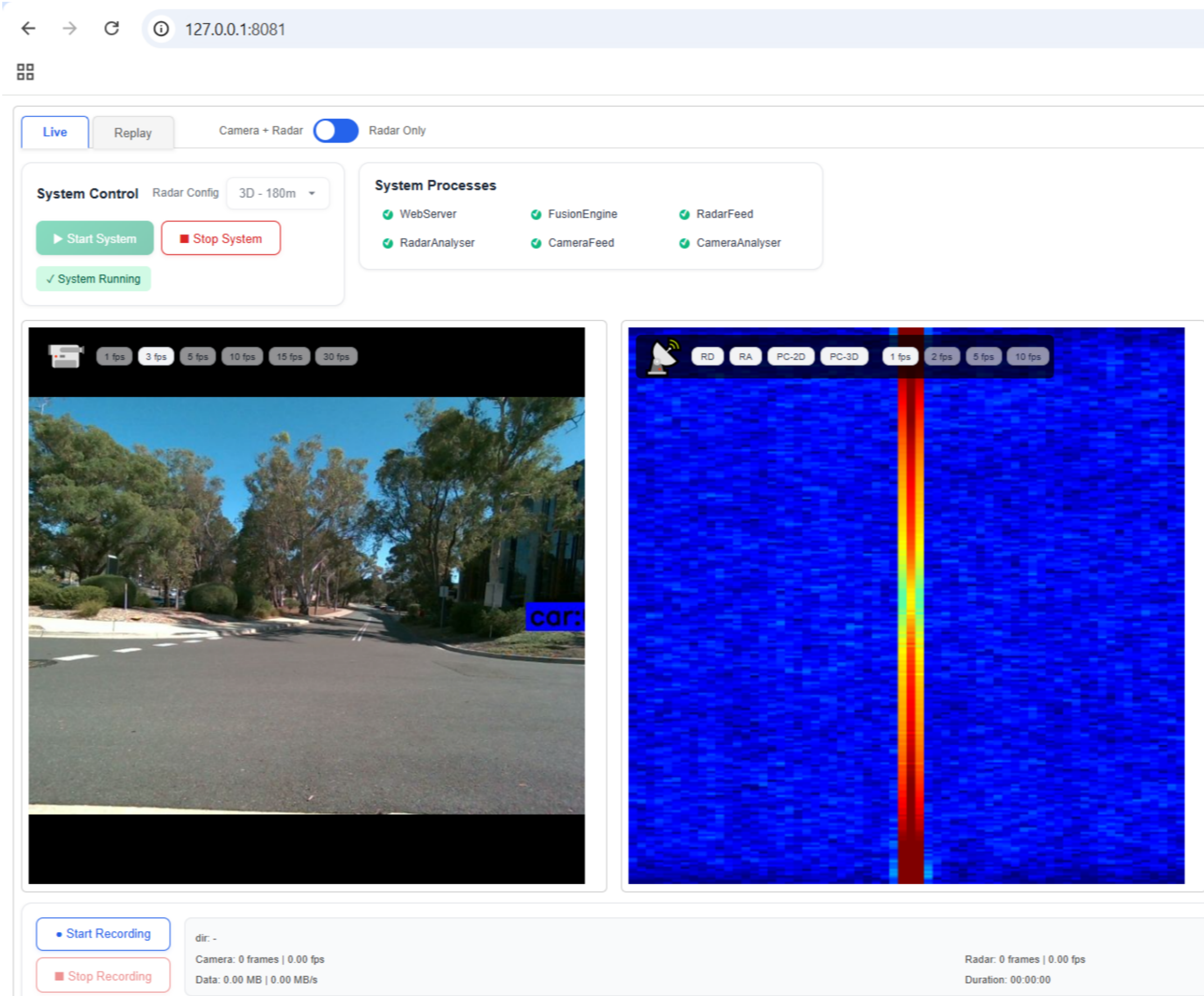


Figure 3: Camera-based object detection interface components: Live Mode.

3 Data Recording and Real-Time Monitoring System: Interface Components and Functionality

This section provides a comprehensive analysis of the user interface for multi-sensor surveillance, with a focus on real-time data recording and investigation. The interface is organized into distinct functional modules that collectively monitor, control, and analyze data from integrated radar and camera sensors. As illustrated in Figures 4 and 3, the system employs visual feedback mechanisms alongside the control elements described below. Each interface component serves a specific role in the operational workflow, from real-time monitoring to post-mission analysis. The subsequent subsections detail these functional options, with Table 1 providing a concise summary of their key elements and primary functions.

3.1 Display and Mode Selection

The interface offers multiple viewing modes to cater to various operational requirements. The **Live** mode enables real-time monitoring of ongoing surveillance activities, displaying current sensor inputs without delay. In this mode, We can access specialized recording controls, sensor configuration options, and real-time performance metrics specific to active surveillance operations. **Replay** mode allows us to review previously recorded data sessions, facilitating post-mission analysis and investigation with comprehensive playback controls. For data visualization, operators can select between **Camera + Radar** mode, which presents fused sensor data for comprehensive situational awareness,

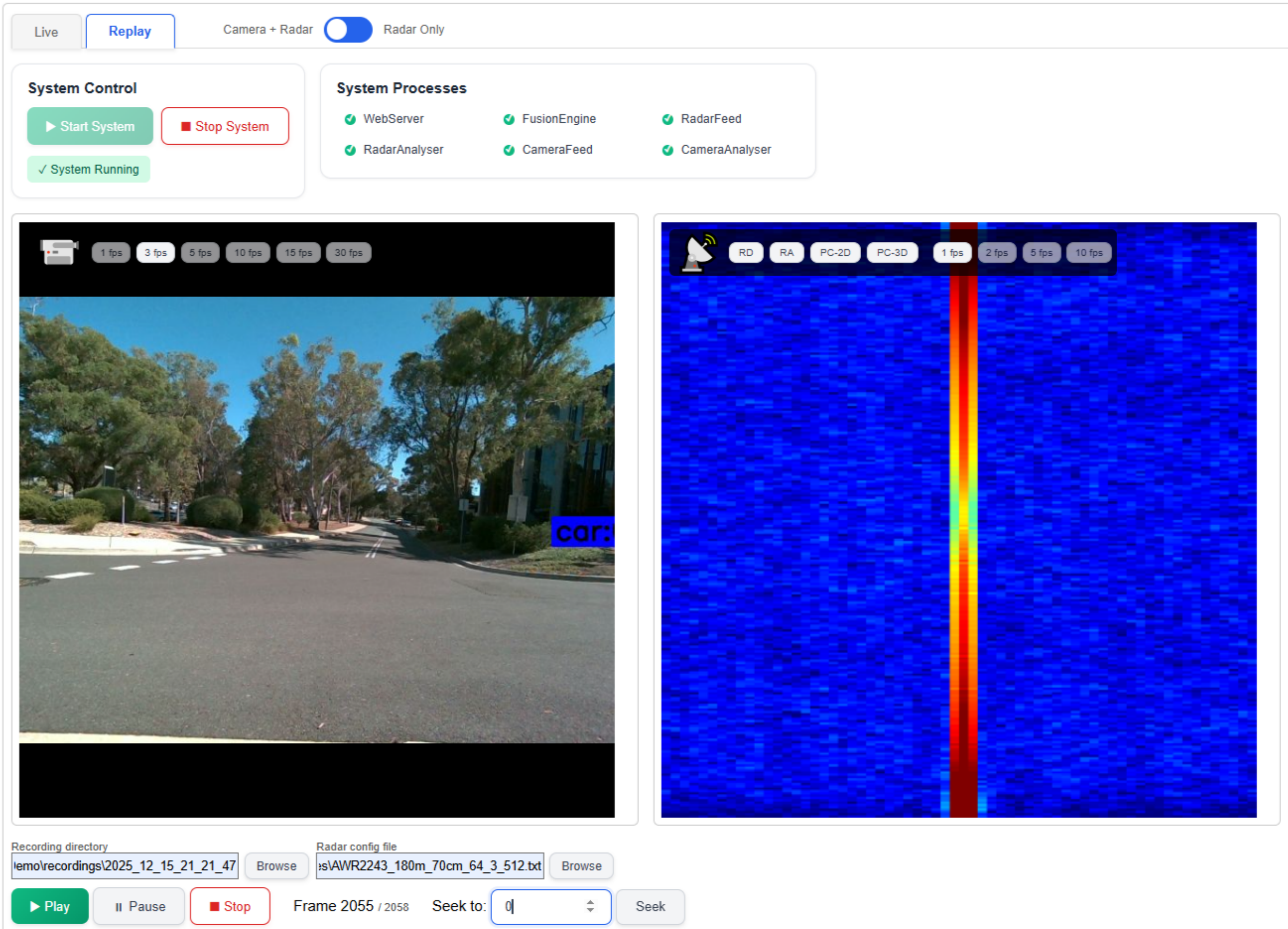


Figure 4: Camera-based object detection interface components: Replay Mode.

or **Radar Only** mode, which isolates radar returns for specialized tracking applications, particularly useful in low-visibility conditions or when visual clutter must be minimized.

3.2 System Control

The core operational controls are mentioned in this section. The **Start System** button initializes all hardware interfaces and software processes, bringing the entire surveillance suite online. Conversely, the **Stop System** command safely terminates all active processes and disengages sensor systems. A persistent **System Status** indicator provides immediate visual confirmation of the system's operational state, typically displaying "System Running" when active, ensuring operators maintain continuous awareness of system availability.

3.3 System Processes

This panel reveals the underlying software architecture, displaying the status of individual computational modules. The **WebServer** component hosts the operational interface itself, enabling remote access and control. The **FusionEngine** performs the critical function of sensor fusion, algorithmically combining radar tracking data with camera imagery to create a unified situational picture. Supporting modules include **RadarFeed** and **CameraFeed**, which manage raw data acquisition from their respective sensors, and **RadarAnalyser** and **CameraAnalyser**, which perform initial processing, object detection, and classification on the incoming data streams.

3.4 Live Mode Recording and Monitoring

During live surveillance operations, the interface provides dedicated controls for immediate data capture. The **Start Recording** and **Stop Recording** buttons enable operators to initiate and terminate data capture sessions during active monitoring without navigating away from the live feed. Real-time performance metrics are displayed during recording operations, including **Camera Statistics** (frame count and frames-per-second), **Data Metrics** (storage volume and data transfer rate), and **Radar Statistics** (frame count and frames-per-second). A **Duration counter** tracks elapsed recording time, providing immediate feedback on the current capture session's length and performance. Additionally, sensor configuration options available during live operations include selection buttons for different operational modes, such as Range-Doppler(**RD**), Range-Azimuth(**RA**), etc, and frame rate controls (**1fps, 2fps, 3fps, 10fps**), allowing us to adjust capture parameters dynamically based on surveillance requirements. In addition, the **Start Recording** and **Stop Recording** buttons enable operators to initiate and terminate data capture sessions during active monitoring without navigating away from the live feed.

3.5 Recording and Configuration

Data management and system customization capabilities are provided in this section. The **Recording Directory** setting specifies the storage location for all captured sessions, with path information displayed for reference. An accompanying **Browse** button enables flexible storage management across different drives or network locations. The **Radar Config File** parameter points to the configuration file containing radar-specific operational parameters such as range settings, sensitivity thresholds, and scanning patterns, with a **Browse** function allowing operators to switch between different pre-configured radar profiles tailored to specific surveillance scenarios.

3.6 Playback Controls

Exclusive to Replay mode, these controls enable precise navigation through recorded data. Standard media transport buttons (**Play, Pause, Stop**) provide basic playback functionality. The **Frame Indicator** displays the current position within a recording relative to the total available frames, offering precise temporal orientation. The **Seek to** function enables direct navigation to specific frame numbers or timestamps, allowing operators to efficiently locate and examine particular events or incidents of interest without manually scanning through entire recordings.

Table 1: System Interface Components Summary

Component Group	Key Elements	Primary Function
Display and Mode Selection	Live, Replay, Camera + Radar, Radar Only	Controls data presentation format and temporal context (real-time vs. recorded)
System Control	Start System, Stop System, System Status	Manages global system operational state and runtime availability
System Processes	Web Server, Fusion Engine, Radar/Camera Feed, Radar/Camera Analyser	Monitors software services responsible for data acquisition, processing, and sensor fusion
Live Mode Recording	Start Recording, Stop Recording, Real-time metrics, Sensor configuration options	Enables immediate data capture during live operation with performance feedback
Recording and Configuration	Recording Directory, Radar Config File (with Browse)	Manages data storage parameters and sensor configuration settings
Playback Controls	Play/Pause/Stop, Frame Indicator, Seek To	Provides navigation and inspection tools for recorded data analysis

4 Radar Frame Processing Pipeline

The radar frame processing pipeline transforms raw analog-to-digital converter (ADC) measurements acquired from a millimetre-wave frequency-modulated continuous-wave (FMCW) radar into geometric representations suitable for perception and sensor fusion. In our pipeline, raw radar frames are processed through a sequence of signal processing stages that progressively extract range, velocity, and angular information [1, 2, 3]. These stages culminate in the generation of either dense or sparse three-dimensional point cloud representations that describe the surrounding environment. The processing pipeline is designed to support both offline high-fidelity analysis and real-time operation. Offline processing prioritizes accuracy and completeness and is aligned with established preprocessing formats such as K-Radar [2], whereas real-time processing emphasizes computational efficiency. In this section, we focus on the full, high-resolution radar frame processing pipeline.

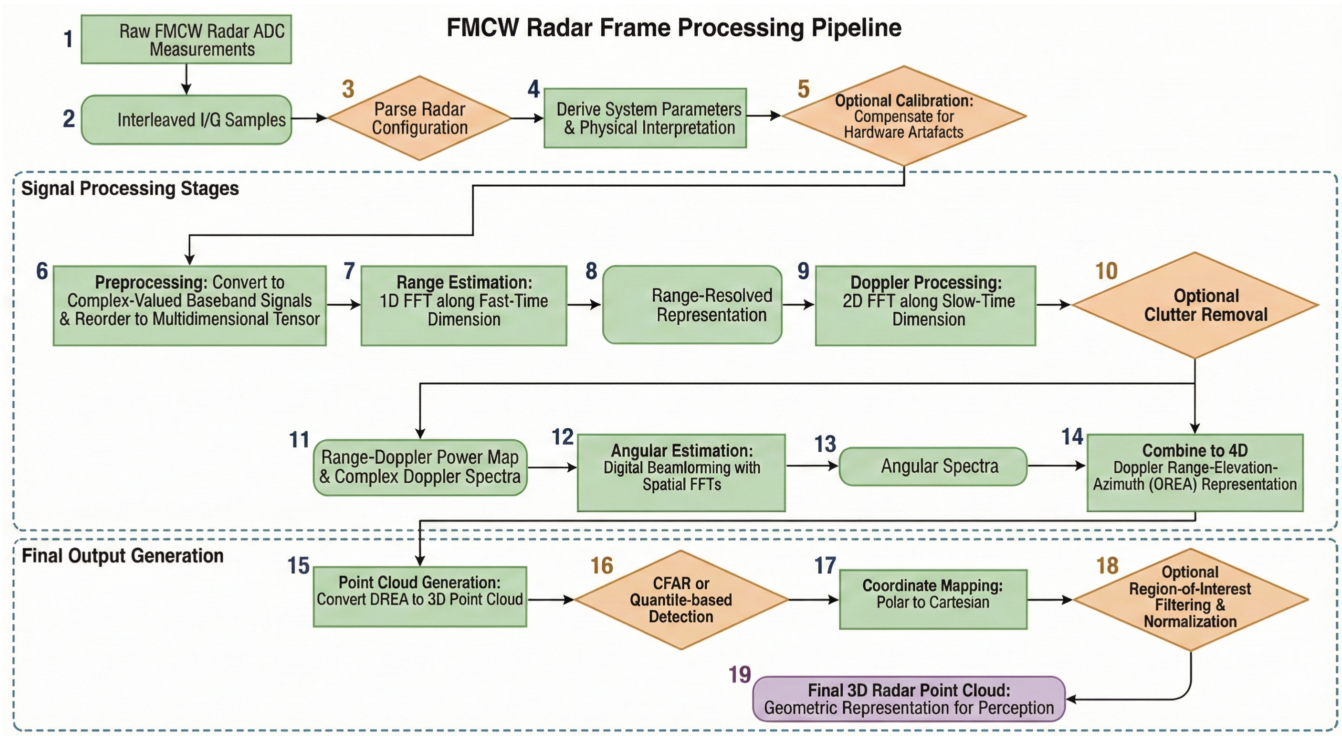


Figure 5: The radar signal processing flow from raw input to 3D point cloud output.

Each radar frame consists of interleaved in-phase (I) and quadrature (Q) ADC samples collected across multiple temporal and spatial dimensions. Samples are acquired over successive chirps, multiple transmit antennas, multiple receive antennas, and fast-time samples within each chirp. For the radar configuration considered in this work, the raw data can be represented as a multidimensional tensor whose dimensions correspond to the number of chirps, transmit antennas, receive antennas, fast-time samples, and I/Q components. The raw samples are stored as 16-bit unsigned integers and transmitted without prior signal conditioning. Before signal processing begins, the radar configuration used during data acquisition is parsed to extract system parameters. These include directly specified parameters such as the number of chirps, active antenna channels, sampling rate, frequency slope, and ADC sample count. From these parameters, additional quantities such as maximum unambiguous range, range resolution, maximum Doppler velocity, and Doppler resolution are derived analytically. This step establishes a consistent physical interpretation of the signal dimensions and ensures that subsequent spectral processing stages yield physically meaningful results.

The preprocessing stage converts the raw interleaved I/Q ADC samples into complex-valued baseband signals. This process involves separating the in-phase and quadrature components, converting the samples to signed values, and forming complex samples of the form $x[n] = I[n] + jQ[n]$. The samples are then reordered into a structured multidimensional tensor indexed by chirps, transmit antennas, receive antennas, and fast-time samples. The resulting complex frame preserves the full spatiotemporal structure of the radar measurements and serves as the input to the subsequent signal processing stages. Calibration steps are optionally applied to compensate for systematic hardware artefacts introduced by the analogue front-end and antenna system. These artefacts typically include constant offsets and channel-dependent amplitude or phase mismatches. Calibration improves consistency across receive channels and enhances signal coherence. For computational efficiency, the chirp and transmit antenna dimensions may be flattened, but the underlying data dimensionality remains unchanged.

Range estimation is performed by applying a one-dimensional Fast Fourier Transform (FFT) along the fast-time dimension of the complex signal. In FMCW radar systems, the beat frequency of the received signal is directly related to target distance. The range FFT maps these frequencies to discrete range bins, producing a range-resolved representation of the scene for each chirp and virtual antenna. Velocity information is extracted through Doppler processing, which applies a second FFT along the slow-time, or chirp, dimension. This operation converts phase evolution across chirps into Doppler frequency components that correspond to the radial velocities of the target. Optional clutter removal may be applied to suppress reflections from stationary objects. The Doppler processing stage produces both a range–Doppler power map and a set of complex Doppler spectra for each virtual antenna, which are required for angular estimation. An FFT shift is typically applied to centre the zero-velocity bin.

Angular information is recovered by exploiting the spatial diversity of the virtual antenna array formed by the transmit and receive elements. Conventional digital beamforming techniques are employed, using FFTs across the spatial dimensions of the array to estimate azimuth and elevation angles. Zero-padding may be used to increase angular bin density for downstream processing, although the fundamental angular resolution remains limited by the physical size and geometry of the antenna array. The resulting angular spectra are combined with range and Doppler dimensions to form a four-dimensional Doppler–Range–Elevation–Azimuth (DREA) representation of the radar data.

In the final stage, the high-dimensional DREA representation is converted into a three-dimensional point cloud. This is achieved by identifying significant reflections using statistical detection methods such as quantile-based selection or constant false alarm rate (CFAR) processing. Detected responses are mapped from polar radar coordinates into a Cartesian coordinate system. Each point is represented by its spatial position, reflected signal power, and radial velocity. Optional region-of-interest filtering and normalization are applied to produce the final radar point cloud, which serves as a compact and informative representation for downstream perception and sensor fusion tasks.

Figure 6: Camera-based object detection.

In Summary, as illustrated in Figure 5, the radar frame processing pipeline begins by ingesting raw, interleaved I/Q ADC measurements and parsing the radar configuration to derive essential system parameters and physical interpretations. These raw samples undergo preprocessing to form complex-valued baseband signals arranged in a multidimensional tensor, where optional calibration is applied to correct hardware artefacts like amplitude or phase mismatches. The data then progresses through a sequence of spectral processing stages: a Range FFT is applied along the fast-time dimension to map target distances, followed by a Doppler FFT along the slow-time dimension to extract radial velocity (with optional clutter removal), and finally, digital beamforming via spatial FFTs is used to estimate azimuth and elevation angles. These stages culminate in a four-dimensional DREA representation, which is converted into a final 3D point cloud through statistical detection (such as CFAR), coordinate transformation from polar to Cartesian, and region-of-interest filtering.

5 Camera Versus Radar for Object Detection in Safe Driving

Reliable object detection is a fundamental requirement for safe driving systems and plays a critical role in enhancing road safety. Camera and radar sensors are widely used to perceive the driving environment and detect road users such as vehicles, pedestrians, and cyclists. Each sensing modality provides complementary capabilities, but also exhibits inherent limitations [4].

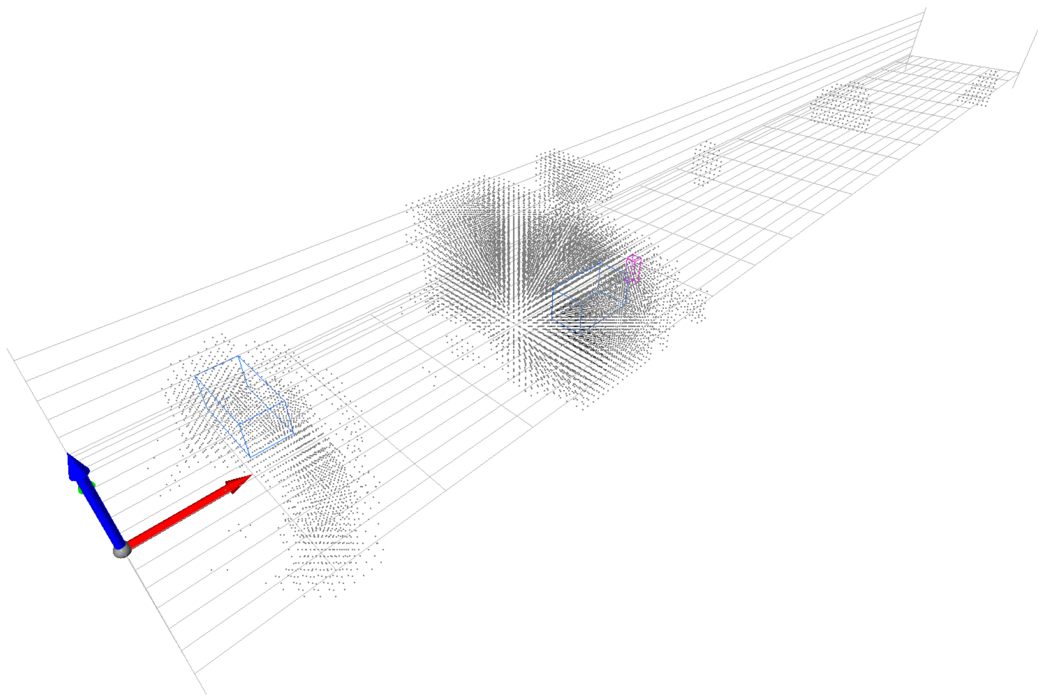
Camera-based object detection offers high spatial resolution and rich visual information, enabling accurate object classification, scene understanding, and recognition of road infrastructure such as lane markings and traffic signs. These capabilities are essential to support safe driving decisions. However, camera performance significantly degrades under adverse environmental conditions, including low illumination, glare, fog, rain, and snow. In such scenarios, reduced image quality can lead to missed detections or false alarms, thus increasing safety risks. In addition, camera-based systems face challenges in detecting distant objects, as visual features become less distinct at long ranges. Occlusion further limits camera performance, particularly in dense traffic environments, where objects may be partially or fully hidden behind other vehicles, pedestrians, or roadside structures. These factors reduce the reliability of vision-only perception in complex road scenes [4, 5]. It can be observed from Figure 6 that the camera-based model successfully detects nearby and clearly visible objects; however, it fails to detect distant objects and those that are partially or fully occluded. For instance, while the nearby vehicles are correctly identified, the pedestrian and the vehicle located farther away are not detected.

Radar-based object detection mitigates many of these limitations by providing direct measurements of object range and relative velocity and by maintaining robust performance under poor lighting and adverse weather conditions. Radar is especially effective for detecting distant and partially occluded objects and for tracking moving targets in low-visibility environments. Nevertheless, radar generally offers lower spatial resolution and limited semantic information compared to camera sensors [4, 6]. Figures 7 and 8 demonstrate the ability of radar-based detection models to identify distant objects and occluded targets, such as a far vehicle and a pedestrian located behind a car, which are not detected by camera-based models. For safe driving and road safety applications, the limitations of camera-based perception highlight the importance of complementary sensing strategies. Camera–radar sensor fusion leverages the strengths of both modalities, improving detection robustness, extending sensing capability, and enhancing overall safety across a wide range of driving conditions.

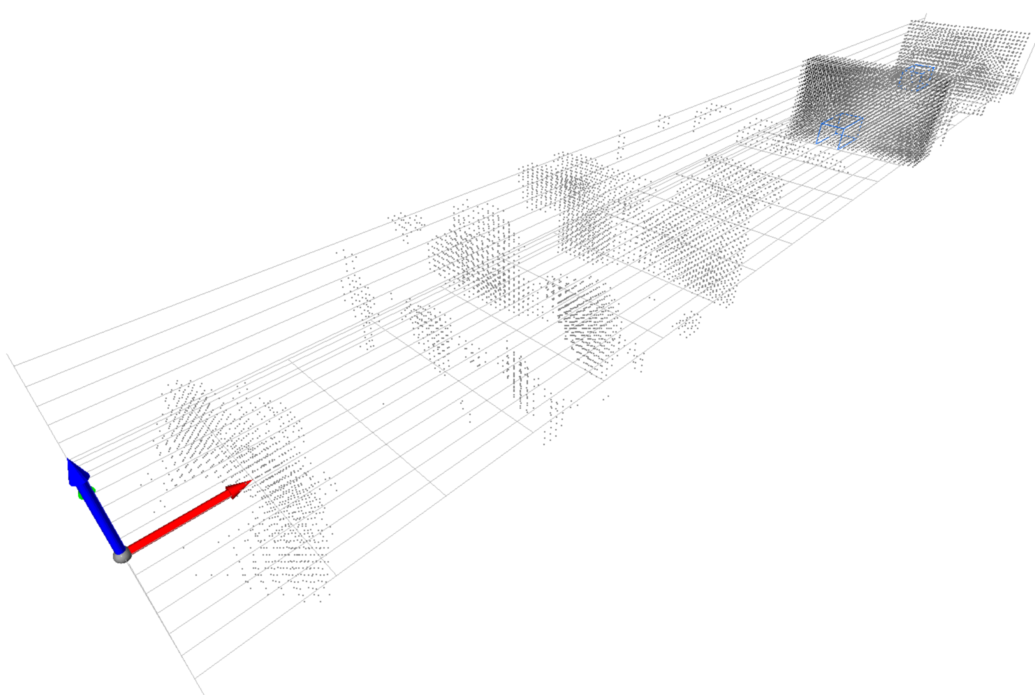
6 Camera–Radar Fusion for Robust 3D Object Detection

The Camera–Radar Fusion model is designed to detect objects in three dimensions by combining information from a camera and a radar sensor. The camera provides rich visual details such as color, texture, and shape, while the radar provides accurate spatial information in the form of three-dimensional points with associated signal power. By using both sensors together, the model benefits from the strengths of each and becomes more reliable in difficult conditions such as low light or partial occlusion [7, 8].

The processing begins with separate feature extraction for each sensor. The camera image is passed through a convolutional neural network that extracts multi-scale visual features representing objects at different sizes. At the same time, the radar data, represented as points with coordinates and power values, is processed by a radar encoder that converts the sparse measurements into structured spatial features. These radar features preserve geometric information that is important for estimating object position and size. After feature extraction, the system aligns the camera and radar information in a common spatial reference. This alignment step ensures that features from both sensors correspond to the same physical locations in the scene. The aligned features are then fused using a combination of projection and attention-based operations, allowing the model to emphasize complementary information from each modality. Camera features contribute semantic understanding, while radar features reinforce geometric accuracy.



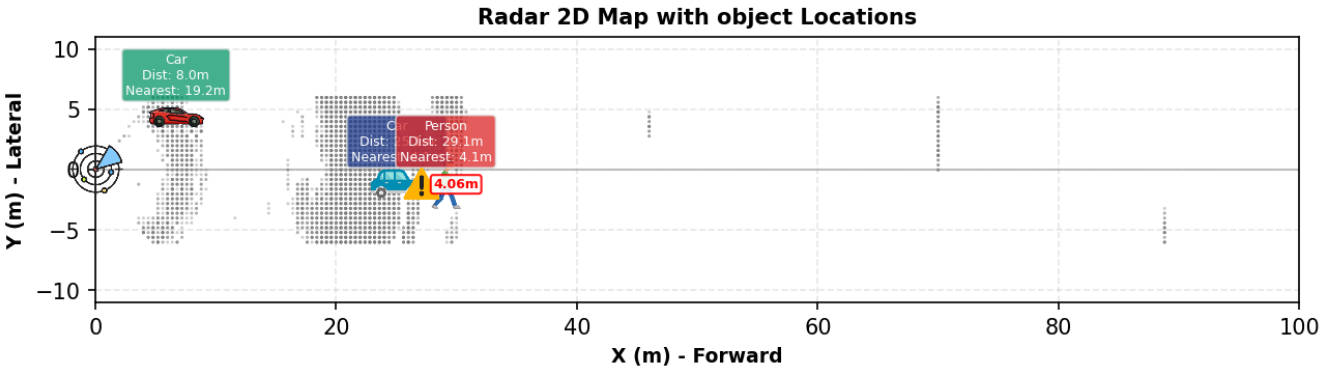
(a) Detection of occluded objects in a 3D radar point cloud.



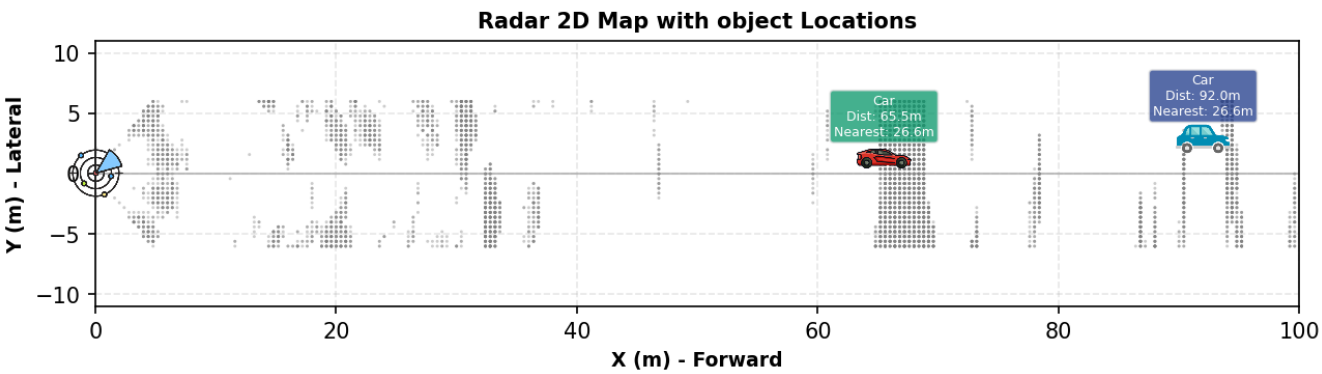
(b) Detection of distant objects in a 3D radar point cloud.

Figure 7: Radar-based object detection from a 3D radar point cloud.

The fused representation is processed by a bird's-eye-view (BEV) encoder, which captures spatial relationships across the ground plane. This representation simplifies reasoning about object layout and interactions by providing



(a) Detection of occluded objects using a Radar-based model.



(b) Detection of distant objects using a Radar-based model.

Figure 8: Radar-based object detection results.

a consistent top-down view of the scene. On top of the BEV features, a detection head predicts object locations, sizes, orientations, and categories. The output consists of 3D bounding boxes that describe each detected object in terms of position, dimensions, and heading. Overall, this architecture combines visual and geometric cues in a unified framework. By fusing camera and radar data at the feature level, the system achieves more stable and accurate 3D object detection than single-sensor approaches, especially in challenging environments where one sensor alone may be unreliable.

Figure 9 details a sequential pipeline for 3D object detection that begins with multi-sensor input from a camera and a radar, processed in parallel along two dedicated branches. The camera's RGB image first passes through a backbone CNN—such as ResNet, Swin Transformer, or ConvNext—to produce hierarchical, multi-scale feature maps (labeled C3, C4, C5) that capture visual details at different resolutions and semantic levels. Simultaneously, raw radar data, typically a sparse point cloud, is encoded by a radar-specific network like PointNet or VoxelNet, which transforms it into a structured spatial feature representation suitable for downstream fusion. This stage ensures that each sensor modality is processed by a network tailored to its inherent data structure before integration.

The pipeline then proceeds to the feature alignment and fusion module, where the two distinct feature sets are spatially and semantically combined. This module performs geometric projection to map the perspective camera features into a common spatial frame with the radar features, followed by cross-attention mechanisms that enable features from one modality to inform and refine the other. The fused output is subsequently passed through a BEV feature encoder—composed of 2D CNN or transformer blocks—that further refines the combined representation. Finally, a 3D detection head takes this encoded feature map and executes multiple tasks in parallel: it predicts object centers via a heatmap, regresses precise 3D bounding box parameters (position, size, and yaw orientation), and assigns class scores. The pipeline concludes by outputting a set of final 3D objects with fully attributed position, size, and heading, ready for use in perception-driven applications. In the next stage of our project, we will focus on

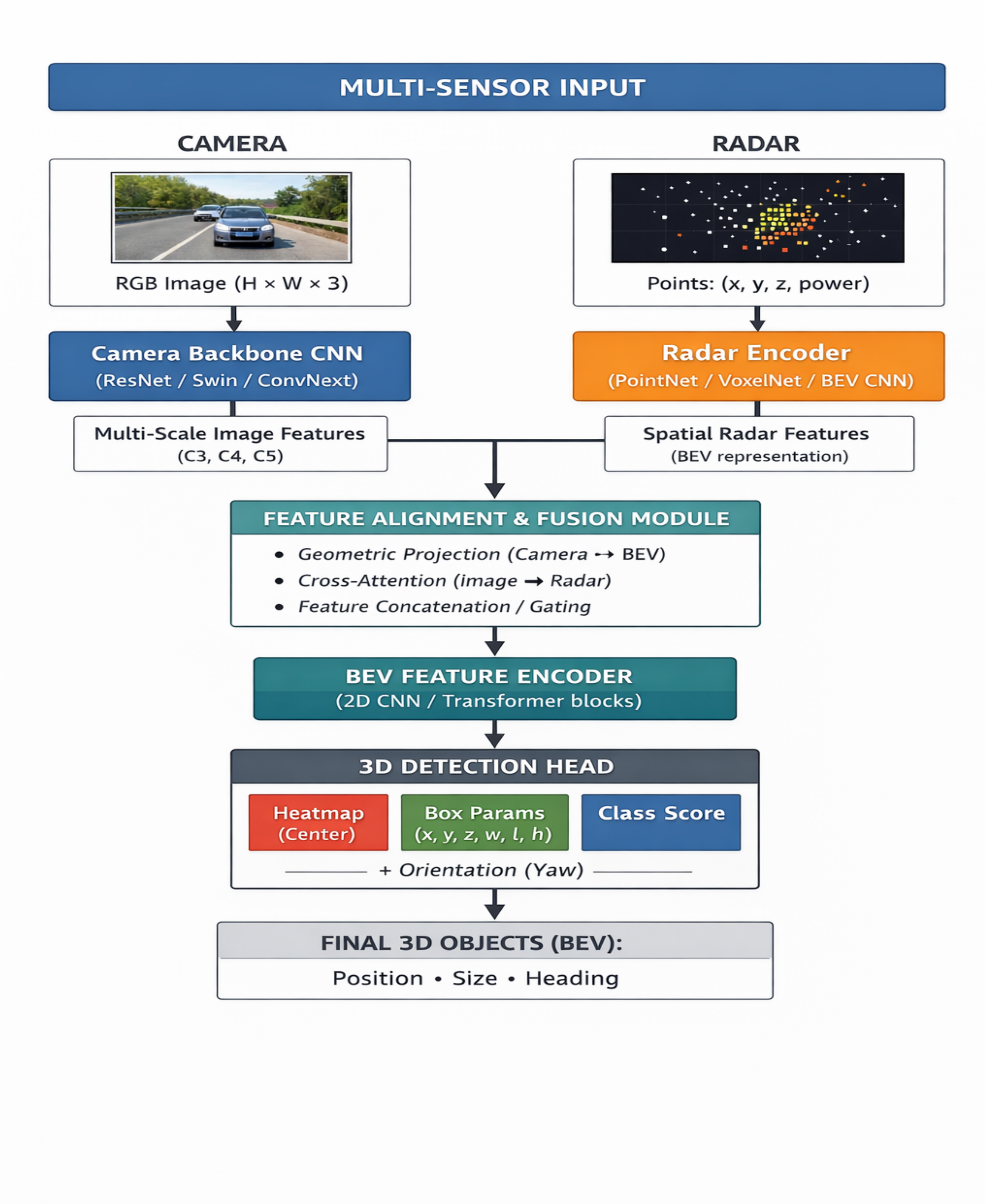


Figure 9: Multi-sensor fusion pipeline for 3d object detection.

developing an advanced camera–radar fusion AI model that leverages the complementary strengths of both sensors to enable robust 3D object detection for enhanced road safety.

7 Risk and Safety Analysis

Risk and safety analysis is a critical component of intelligent driving systems because it quantifies the threat level of surrounding objects and supports timely and reliable decision making. In camera–radar perception pipelines, risk estimation is derived from 3D object detection outputs, including object position, distance, relative velocity, and motion direction. These parameters are used to compute safety indicators such as minimum safe distance, time-to-collision (TTC), and lane or path conflict, which are widely adopted in autonomous driving research [9, 10].

Among these metrics, TTC provides a simple and intuitive estimate of how soon a collision may occur if current motion patterns remain unchanged. However, real driving environments often contain ambiguous situations caused by occlusion, sensor noise, or adverse weather. For example, a pedestrian partially hidden behind a parked vehicle may be weakly visible in camera images while being clearly detected by radar. Similarly, in heavy rain or low-light conditions, camera confidence may decrease while radar continues to provide stable range and velocity measurements. In such cases, single-sensor risk estimation becomes unreliable. Camera–radar fusion reduces this ambiguity by combining complementary sensor information, leading to more accurate distance and velocity estimation and more reliable risk prediction [11, 12].

Recent studies also emphasize the importance of uncertainty-aware risk assessment, where perception confidence is incorporated into the risk score to ensure safer decision making under ambiguous conditions [13, 14, 15]. By integrating these risk metrics into the perception pipeline, autonomous systems can prioritize dangerous objects, generate early warnings, and support safer driving behavior. Therefore, risk and safety analysis serves as a crucial link between perception and decision-making in road-safety applications.

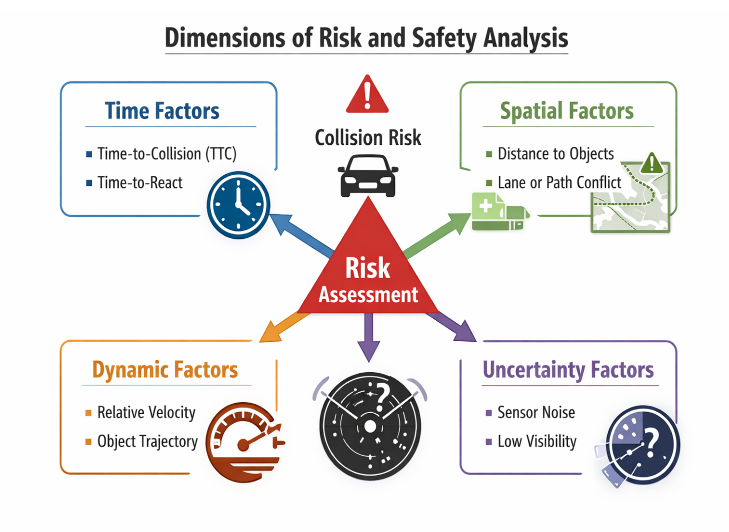


Figure 10: Key dimensions of risk and safety analysis for collision risk estimation in intelligent driving systems.

Figure 10 illustrates the main dimensions involved in risk and safety analysis for intelligent driving systems, centered around collision risk as the primary safety objective. Risk assessment integrates four complementary factors: time, spatial, dynamic, and uncertainty dimensions. The time dimension includes metrics such as time-to-collision and time-to-react, which indicate how urgently a response is required. The spatial dimension captures the geometric relationship between the ego vehicle and surrounding objects, including distance and lane or path conflicts. The dynamic dimension describes motion-related properties such as relative velocity and predicted object trajectories, which determine how the situation is likely to evolve. Finally, the uncertainty dimension accounts for sensor noise, occlusions, and low-visibility conditions that reduce perception reliability. By combining these four dimensions, the system produces a comprehensive and robust risk assessment that supports early warning generation, safer decision making, and improved road-safety performance.

8 Challenges and Future Plan

3D object deection for road safety under adverse weather conditions using camera-radar fusion. Reliable 3D object detection is critical for road safety, especially under adverse weather conditions such as rain, fog, or low light. Cameras provide rich visual detail but struggle with distant, occluded, or poorly lit objects, while radar offers robust range and velocity measurements but limited semantic information. Individually, each sensor has limitations, highlighting the need for fusion. Our next project stage will focus on developing an advanced camera–radar fusion AI model for 3D detection under challenging conditions. The model will combine complementary sensor features, address occlusions and alignment issues, and operate in real time to ensure accurate detection of all relevant objects, ultimately improving road safety across diverse environments.

Cascaded radar and stereo camera for 3d detection. Integrating cascaded radar with stereo cameras enables robust 3D perception for road safety. Stereo cameras provide high-resolution depth information, while cascaded radar—using multiple radar layers or units—offers enhanced spatial resolution, wider coverage, and better detection of occluded or distant objects compared to a single radar. Together, they complement each other, providing reliable detection even in poor visibility, occlusions, or complex traffic scenarios. Our plan is to develop a cascaded radar–stereo camera system where both sensors operate in parallel, each contributing its strengths. Cascaded radar provides accurate range, velocity, and spatial information, while the stereo camera offers detailed depth and shape information. This combination improves detection accuracy, handles occlusions, reduces blind spots, and ensures real-time 3D perception, enhancing road safety under diverse environmental conditions.

Benchmarking radar datasets: challenges and solution. Evaluating radar datasets for 3D object detection presents several key challenges. Hardware diversity is a major issue, as datasets use different radar types, including 360° radars, commercially available 3D or 4D radars, and development radars, resulting in incompatible data formats, antenna configurations, and signal processing requirements. Missing configuration details further complicate research, as many datasets do not provide exact radar parameters, forcing reliance on approximations that reduce model performance. Additionally, incomplete preprocessing pipelines—from raw ADC data to model input—limit reproducibility and prevent accurate replication of reported results. Together, these challenges make benchmarking and development across multiple datasets difficult. We plan to record a comprehensive radar dataset with well-defined and rigorous radar configurations and preprocessing pipelines. By ensuring consistent hardware setup, documented configurations, and a complete signal processing pipeline from raw data to model input, the dataset will enable reproducible research and accurate evaluation of radar-based 3D detection models.

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